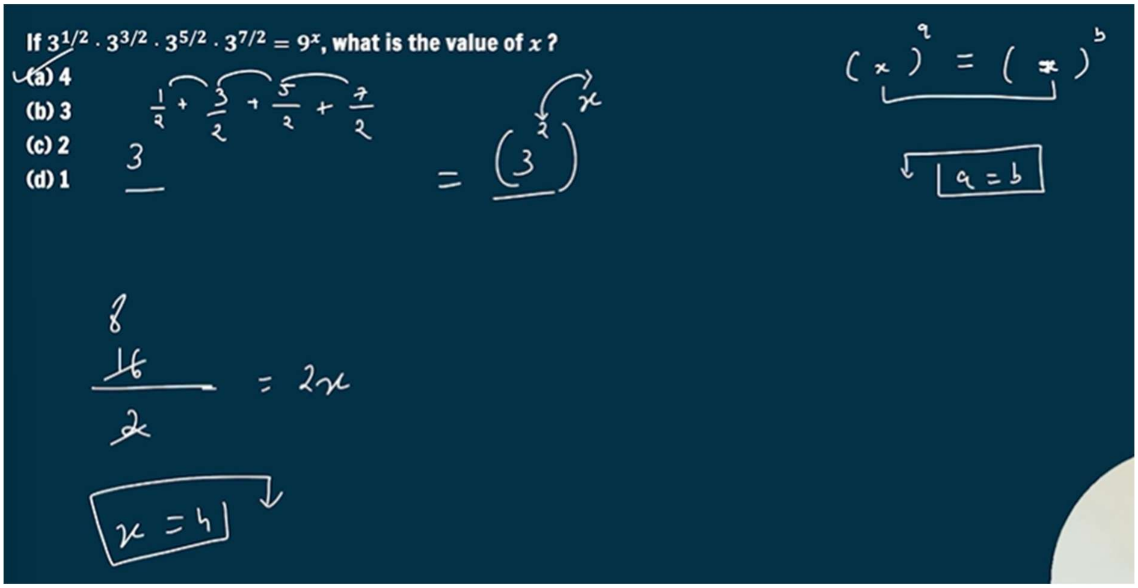
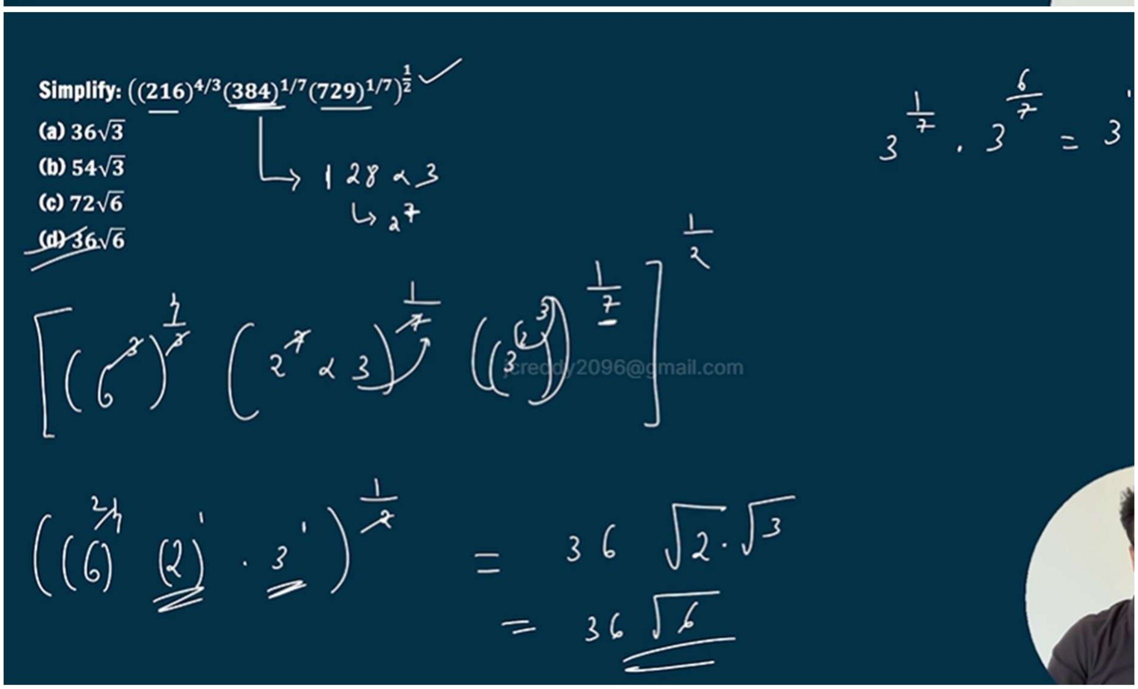
→Day-4

→Simplification

If bases equal we can equate powers





Simplify $\frac{2^{m+2}6^{m+2}10^{2m+2}15^{m+1}25^{m-n}}{3^{2m-3}4^{2m}6^{6}5^{5m+2n}25^{1-2n}}$.

(a) 1

(b) $2^m3^{m-1}5$

(c) 5^{2n}

(d) 5

$2 \cdot 3 \qquad 5 \cdot 2$

$$= \frac{2^{m+2} \cdot 2^{m+2} \cdot 3^{m+2} \cdot 2^{2m+2} \cdot 5^{2m+2} \cdot 5^{m+1} \cdot 3^{m+1} \cdot 5^{2m-2n}}{3^{2m-3} \cdot 2^{4m} \cdot 2^{6} \cdot 3^{6} \cdot 5^{5m+2n} \cdot 5^{2-4n}}$$

$$2^{\overset{0}{4m+6-4m-6}} \cdot 3^{\overset{0}{2m+3-2m+3-6}} \cdot 5^{\overset{1}{5m+3-2n-5m-2n-2+4n}}$$

$$= 5$$

Simplify : $x^{\frac{a}{a+b+c}} \cdot x^{\frac{b}{a+b+c}} \cdot x^{\frac{c}{a+b+c}}$.

(a) 1

(b) x

(c) 0

(d) x^{abc}

$$= x^{\frac{a+b+c}{a+b+c}} = x^{1} = x$$

If $5^{1/a} = 7^{1/b} = 35^{1/c}$ find c in terms of a and b.

(a) $\frac{ab}{a+b}$

(b) $a + b$

(c) $\frac{ab}{a-b}$

(d) $a - b$

$$5^{\frac{1}{a}} = 7^{\frac{1}{b}} = (7 \cdot 5)^{\frac{1}{c}} = k$$

$$5 = k^{a}, \quad 7 = k^{b}, \quad (7 \cdot 5) = k^{c}$$

$$k^{a} \cdot k^{b} = k^{c} = k^{a+b} \Rightarrow \boxed{c = a + b}$$

$$\frac{a^5}{a^7} = a^{5-7} = a^{-2}$$

$$a^{-2} = a^{1/2}$$

$$\downarrow$$

$$\frac{1}{a^2} \neq \sqrt{a}$$

Simplify $\left(\frac{a^3b^6c^9}{a^4b^8c^{12}}\right)^9 \div \left(\frac{a^2b^4c^6}{a^3b^7c^5}\right)^8$.

(a) ab^6c^{35}

~~(b)~~ $\frac{b^6}{ac^{35}}$

(c) $\frac{ac^{35}}{b^6}$

(d) $\frac{c^{35}}{ab^6}$

$$\left(a^{-1}\cdot b^{-2}\cdot c^{-3}\right)^9$$

$$= \frac{\left(\frac{1}{a}\cdot\frac{1}{b^2}\cdot\frac{1}{c^3}\right)^9}{\left(\frac{1}{a}\cdot\frac{1}{b^3}\cdot c\right)^8}$$

$$\frac{\frac{1}{b^{18}}}{\frac{1}{b^{24}}}$$

$$b^6 = \frac{1}{a}\cdot b^6\cdot\frac{1}{c^{35}} = \frac{b^6}{a\cdot c^{35}}$$

Given $3^{4x+2} = 729$ and $2^{3y+1} = 8192$, find $\frac{2y+x}{y-x}$. $\rightarrow$ $\frac{8+1}{4-1} = \frac{9}{3}$ $= 3$

$2^7 = 128$

(a) 2
(b) 1
(c) 4
(d) 3

$3^{4x+2} = (3^2)^3$

$4x + 2 = 6$

$\boxed{x = 1}$

$2^{3y+1} = 2^{13}$

$3y + 1 = 13$

$\boxed{y = 4}$

Given $5^a + 2^{b+1} = 189, 5^{a+1} + 2^{b-2} = 633$. If a and b are positive integers find a + b. $= 8$

(a) 8
(b) 7
(c) 10
(d) 9

$5^a + 2^{b+1} = 189$

$125 + 64 = 189$

$\boxed{5^3 + 2^6 = 189}$

$a = 3, \ b + 1 = 6$

$\boxed{b = 5}$

$5^{a+1} + 2^{b-2} = 633$

$5^4 + 2^3 = 625 + 8 = 633$

$\left(\sqrt[3]{\frac{3}{5}}\right)^{3x+12} = \left(\sqrt[4]{\frac{25}{9}}\right)^{3x+1}$. **What is the value of x ?**

(a) $-\frac{17}{7}$

(b) $-\frac{9}{5}$

(c) 7

(d) -15

$$\left(\frac{3}{5}\right)^{\frac{3x+12}{3}} = \left(\frac{25}{9}\right)^{\frac{3x+1}{4}}$$

$$\sqrt[3]{x^1} = x^{\frac{1}{3}}$$

$$\sqrt[4]{x^1} = x^{\frac{1}{4}}$$

$$\sqrt[n]{x^a} = (x)^{\frac{a}{n}}$$

$$= \left(\frac{5}{3}\right)^{2\times\frac{(3x+1)}{\cancel{4}\,2}}$$

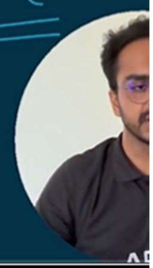
$$= \left(\frac{3}{5}\right)^{-\left(\frac{3x+1}{2}\right)}$$

$$\frac{3x+12}{3} = \frac{-(3x+1)}{2}$$

$$6x+24 = -9x-3$$

$$\cancel{15}\,x = -\cancel{27}\ 9$$

$$x = -9/5$$



Simplify: $\left(3^{\frac{1}{3}}9^{\frac{2}{3}}27^{\frac{4}{3}}81^{\frac{8}{3}}243^{\frac{10}{3}}729^{\frac{12}{3}}\right)^{\frac{2}{19}}$.

(a) 27

(b) 81

(c) 243

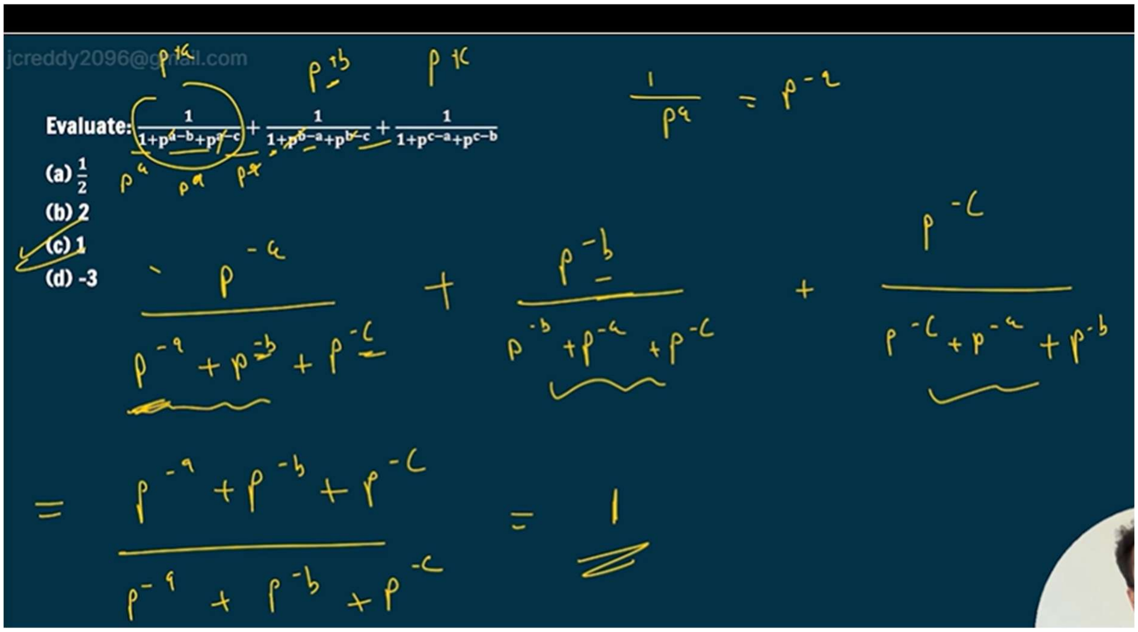
(d) 729

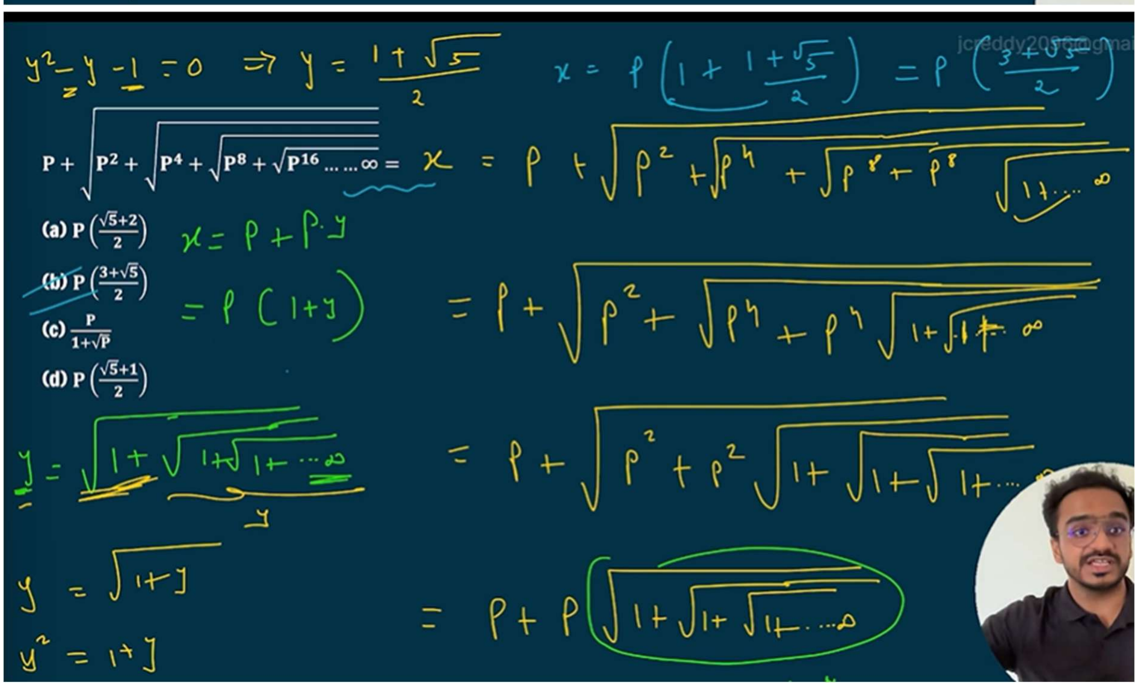
$$\left(3^{\frac{1}{3}}\cdot\left(3^2\right)^{\frac{2}{3}}\cdot\left(3^3\right)^{\frac{4}{3}}\cdot\left(3^4\right)^{\frac{8}{3}}\cdot\left(3^5\right)^{\frac{10}{3}}\cdot\left(3^6\right)^{\frac{12}{3}}\right)^{\frac{2}{19}}$$

$$\left(3^{\frac{1}{3}+\frac{4}{3}+\frac{12}{3}+\frac{32}{3}+\frac{50}{3}+\frac{72}{3}}\right)^{\frac{2}{19}}$$

$$\left(3^{\frac{\cancel{171}\ 57}{\cancel{3}}}\right)^{\frac{2}{19}} = 3^{\cancel{57}\,3\cdot\frac{2}{\cancel{19}}} = 3^6 = 729$$

→Day-5


jcreddy2096@gmail.com
Evaluate: $\frac{1}{1+p^{a-b}+p^{a-c}}+\frac{1}{1+p^{b-a}+p^{b-c}}+\frac{1}{1+p^{c-a}+p^{c-b}}$
(a) $\frac{1}{2}$
(b) 2
(c) 1
(d) -3
$\frac{1}{p^a} = p^{-a}$
$\frac{p^{-a}}{p^{-a}+p^{-b}+p^{-c}} + \frac{p^{-b}}{p^{-b}+p^{-a}+p^{-c}} + \frac{p^{-c}}{p^{-c}+p^{-a}+p^{-b}}$
$= \frac{p^{-a}+p^{-b}+p^{-c}}{p^{-a}+p^{-b}+p^{-c}} = 1$


jcreddy2096@gmail
$y^2 - y - 1 = 0 \Rightarrow y = \frac{1+\sqrt{5}}{2}$
$x = p\left(1 + \frac{1+\sqrt{5}}{2}\right) = p\left(\frac{3+\sqrt{5}}{2}\right)$
$P+\sqrt{P^2+\sqrt{P^4+\sqrt{P^8+\sqrt{P^{16}\ldots\ldots\infty}}}} = x$
(a) $P\left(\frac{\sqrt{5}+2}{2}\right)$
(b) $P\left(\frac{3+\sqrt{5}}{2}\right)$
(c) $\frac{P}{1+\sqrt{P}}$
(d) $P\left(\frac{\sqrt{5}+1}{2}\right)$
$x = P + P\cdot y$
$= P(1+y)$
$y = \sqrt{1+\sqrt{1+\sqrt{1+\ldots\infty}}}$
$y = \sqrt{1+y}$
$y^2 = 1+y$
$= p + \sqrt{p^2 + p^2\sqrt{1+\sqrt{1+\sqrt{1+\ldots\infty}}}}$
$= p + p\left(\sqrt{1+\sqrt{1+\sqrt{1+\ldots\infty}}}\right)$

If $a^2 + b^2 + c^2 = 1$, what is the value of $\left(\sqrt[-bc]{d^a}\right)\times\left(\sqrt[-ca]{d^b}\right)\times\left(\sqrt[-ab]{d^c}\right)$?

(a) $(d)^{1/abc}$

(b) d^{-abc}

(c) $(d)^{abc}$

(d) $d^{-\frac{1}{abc}}$

$$= (d)^{\frac{a}{-bc}} \cdot (d)^{-\frac{b}{ac}} \cdot (d)^{-\frac{c}{ab}}$$

$$= (d)^{-\left(\frac{a}{bc} + \frac{b}{ac} + \frac{c}{ab}\right)}$$

$$= (d)^{-\left(\frac{a^2+b^2+c^2}{abc}\right)}$$

$$= d^{-\frac{1}{abc}}$$

If $x = \frac{1}{\sqrt{2}}\sqrt{\frac{\sqrt{6}+\sqrt{2}}{\sqrt{6}-\sqrt{2}}}$, find the value of $5x^2 - 5x$.

(a) $\frac{\sqrt{5}}{2\sqrt{2}}$

(b) $\frac{1}{2}$

(c) $\frac{5}{4}$

(d) $\frac{5}{2}$

$$x = \frac{1}{\sqrt{2}}\sqrt{\frac{(\sqrt{6}+\sqrt{2})\times(\sqrt{6}+\sqrt{2})}{\underbrace{(\sqrt{6}-\sqrt{2})}_{a-b}\times\underbrace{(\sqrt{6}+\sqrt{2})}_{a+b}}}$$

$$= \frac{1}{\sqrt{2}}\sqrt{\frac{(\sqrt{6}+\sqrt{2})^2}{\underbrace{6-2}_{4}}}$$

$$= \frac{1}{\sqrt{2}}\,\frac{(\sqrt{6}+\sqrt{2})}{2} = \frac{\overset{\sqrt{3}}{\cancel{\sqrt{6}}}}{\cancel{\sqrt{2}}\cdot 2} + \frac{\cancel{\sqrt{2}}}{\cancel{\sqrt{2}}\cdot 2}$$

$$= \frac{\sqrt{3}+1}{2}$$

$$5\left(\frac{\sqrt{3}+1}{2}\right)\left(\frac{\sqrt{3}+1}{2} - 1\right)$$

$$5\left(\frac{\sqrt{3}+1}{2}\right)\left(\frac{\sqrt{3}-1}{2}\right)$$

$$5 \cdot \frac{\cancel{3-1}^{\,1}}{\cancel{4}\,2} = \frac{5}{2}$$

$\frac{5}{3^{2/3}-6^{1/3}+2^{2/3}}$ is equal to

(a) $3^{1/3}+2^{1/3}$

(b) $3^{1/3}-2^{1/3}$

(c) $2^{1/3}-3^{1/3}$

(d) $3^{2/3}+2^{2/3}$

$6 = (2\cdot3)^{\frac{1}{3}}$

$(3)^{\frac{1}{3}} = a, \quad (2)^{\frac{1}{3}} = b$

$(a+b)(a^2-ab+b^2)$

$= a^3+b^3$

$= \frac{5}{(3)^{\frac{2}{3}} - (2)^{\frac{1}{3}}\cdot(3)^{\frac{1}{3}} + (2)^{\frac{2}{3}}} \times \frac{3^{\frac{1}{3}}+2^{\frac{1}{3}}}{3^{\frac{1}{3}}+2^{\frac{1}{3}}}$

$= \frac{5\left(3^{\frac{1}{3}}+2^{\frac{1}{3}}\right)}{\left((3)^{\frac{1}{3}}\right)^3 + \left(2^{\frac{1}{3}}\right)^3} = \frac{\cancel{5}\left(3^{\frac{1}{3}}+2^{\frac{1}{3}}\right)}{\cancel{5}}$

$= 3^{\frac{1}{3}}+2^{\frac{1}{3}}$

If $x-4=\sqrt{11}+\sqrt{7}$, find the value of $x^4-16x^3+60x^2+32x$.

(a) 244

(b) 264

(c) 280

(d) 304

$(x-4)^2 = (\sqrt{11}+\sqrt{7})^2$

$(x^2+16-8x)^2 = (18+2\sqrt{77})^2$

$x^4+64x^2+4-16x^3+32x-4x^2 = 308$

$x^4-16x^3+60x^2+32x = 304$